

Beats

When two sounding bodies of nearly the same frequency and amplitude are sounded together, the resultant sound consists of alternate maxima and minima (as shown in Fig.). This phenomenon which waxing and waning of sound at regular intervals is heard is called beats. The number of beats heard per second is equal to the difference in frequency between the two sounding bodies. However, beats are heard only when the difference in frequency is not more than ten.

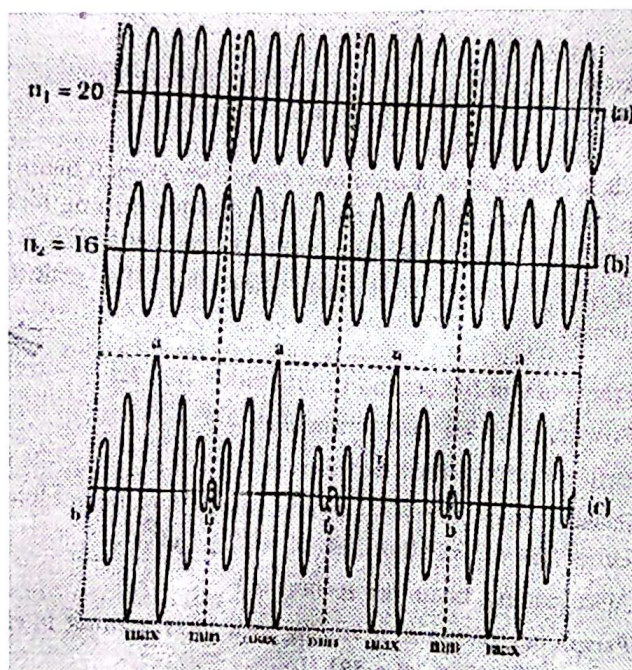


Fig. represents the resultant displacement curve when the notes (a) of frequency 20 hertz and (b) of frequency 16 hertz are sounded simultaneously. From the resultant curve (c) it is clear that in one sec, there are four positions of maximum intensity of sound represented by 'a' and four positions of minimum intensity of sound represented by 'b'. The number of maxima and minima per second is four, which is equal to the difference in frequency between the two notes.

Analytical treatment of beats:

Consider two wave trains of nearly same frequency and amplitudes

$$y_1 = a \sin 2\pi n_1 t \quad \dots\dots\dots (1)$$

$$y_2 = b \sin 2\pi n_2 t \quad \dots\dots\dots (2)$$

$$\begin{aligned}
\therefore y &= y_1 + y_2 = a \sin 2\pi n_1 t + b \sin 2\pi n_2 t \\
&= a \sin 2\pi n_1 t + b \sin 2\pi [n_1 - (n_1 - n_2)]t \\
&= a \sin 2\pi n_1 t + b [\sin 2\pi n_1 t \cos 2\pi (n_1 - n_2)t - \cos 2\pi n_1 t \sin 2\pi (n_1 - n_2)t] \\
&= \sin 2\pi n_1 t [a + b \cos 2\pi (n_1 - n_2)t] - \cos 2\pi n_1 t [b \sin 2\pi (n_1 - n_2)t]
\end{aligned}$$

Let $a + b \cos 2\pi (n_1 - n_2)t = A \cos \theta$

$b \sin 2\pi (n_1 - n_2)t = A \sin \theta$

$\therefore$ the resultant displacement, $y = A \sin (2\pi n_1 t - \theta)$

Then, the resultant amplitude

$$A = \sqrt{a^2 + b^2 + 2ab \cos 2\pi (n_1 - n_2)t}$$

(1) Resultant amplitude is maximum, when

$$\cos 2\pi (n_1 - n_2)t = \pm 1 = \cos 2p\pi$$

$$\text{or, } t = \frac{p}{n_1 - n_2} \text{ (if, } p = 0, 1, 2, 3, \dots \text{)}$$

The maximum intensity of sound will be heard at time instants, $0, \frac{1}{n_1 - n_2}, \frac{2}{n_1 - n_2}, \dots$

(2) Resultant amplitude is minimum, when

$$\cos 2\pi (n_1 - n_2)t = 0 = \cos (2p + 1)\pi / 2$$

$$\text{or, } t = \frac{2p + 1}{2(n_1 - n_2)} \text{ (if, } p = 0, 1, 2, 3, \dots \text{)}$$

The minimum intensity of sound will be heard at time instants, $\frac{1}{2(n_1 - n_2)}, \frac{3}{2(n_1 - n_2)}, \dots$

Thus, the maximum and minimum occur alternately after equal intervals of time $\frac{1}{(n_1 - n_2)}$.

The time interval between two successive maxima or between two successive minima is

$\frac{1}{(n_1 - n_2)}$. Hence, sound will be fluctuated $(n_1 - n_2)$ times/sec. Therefore, the number of beats

produced per second = $n_1 - n_2$.

Interference of Sound Waves

If two or more sound waves travel through the same medium, superposition of the waves takes place. The particles of the medium vibrate according to the vector sum of the displacements due to interfering wave at that instant. As a result of the superposition, the distribution of energy in the medium is different than that of due to individual waves. The resultant vibration of the particles will depend upon the amplitude, time period, phase difference and direction of the interfering waves.

Conditions for interference of sound waves:

- (i) The two sound source must be coherent i.e., emit wave of same frequency and amplitude.
- (ii) Phase difference between waves must remain constant.
- (iii) Wave trains must travel in same direction.

Analytical treatment of interference:

Consider two simple harmonic waves of H. same frequency and amplitude

$$y_1 = a \sin \frac{2\pi}{\lambda} (vt - x)$$

$$y_2 = a \sin \frac{2\pi}{\lambda} [(vt - x) + \phi]$$

$$\begin{aligned} \therefore y = y_1 + y_2 &= a \sin \frac{2\pi}{\lambda} (vt - x) + a \sin \frac{2\pi}{\lambda} (vt - x) \cos \phi + a \cos \frac{2\pi}{\lambda} (vt - x) \sin \phi \\ &= \left[\sin \frac{2\pi}{\lambda} (vt - x) \right] (a + a \cos \phi) + \left[\cos \frac{2\pi}{\lambda} (vt - x) \right] a \sin \phi \dots\dots\dots(i) \end{aligned}$$

Let $a + a \cos \phi = A \cos \theta \dots\dots\dots(ii)$

$a \sin \phi = A \sin \theta \dots\dots\dots(iii)$

$$\begin{aligned} \therefore A^2 &= a^2 + 2a^2 \cos \phi + a^2 \cos^2 \phi + a^2 \sin^2 \phi \\ &= 2a^2 + 2a^2 \cos \phi \\ &= 2a^2 (1 + \cos \phi) \\ &= 2a^2 \cdot 2 \cos^2 \phi / 2 \end{aligned}$$

$\therefore I = 4a^2 \cos^2 \phi / 2 \dots\dots\dots(iv)$

Eqⁿ (iv) represents the intensity of resultant wave. It is minimum, when

$$\cos \varphi / 2 = 0 = \cos(2n + 1)\pi / 2$$

$$\therefore \varphi = (2n + 1)\pi, \quad \text{If, } n = 0, 1, 2, 3, \dots \text{ then, } \varphi = \pi, 3\pi, 5\pi, \dots$$

i.e., the phase difference is $(2n + 1)\pi$, or the path difference is $(2n + 1)\frac{\lambda}{2}$.

It is maximum, when

$$\cos \varphi / 2 = \pm 1 = \cos n\pi$$

$$\varphi = 2n\pi, \quad \text{If, } n = 0, 1, 2, 3, \dots \text{ then, } \varphi = 0, 2\pi, 4\pi, \dots$$

i.e., the phase difference is $2n\pi$, or the path difference is $n\lambda$

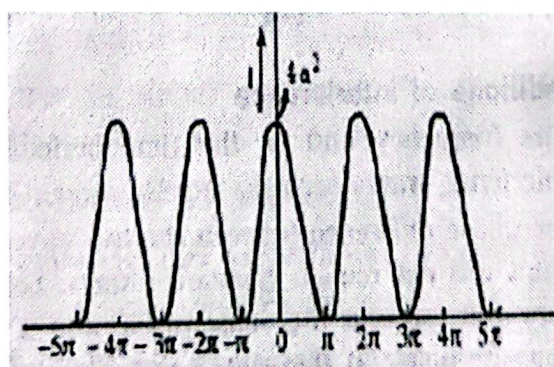
Putting the values of (ii) & (iii) in Eqⁿ (i), we see

$$\begin{aligned} y &= \left[\sin \frac{2\pi}{\lambda} (vt - x) \right] A \cos \theta + \left[\cos \frac{2\pi}{\lambda} (vt - x) \right] A \sin \theta \\ &= A \sin \left[\frac{2\pi}{\lambda} (vt - x) + \theta \right] \end{aligned}$$

This is the simple harmonic motion.

Energy distribution due to interference:

From equation (iv), the intensity at the maxima is $4a^2$ and the intensity at the minima is zero. According to the law of conservation of energy, the energy cannot be destroyed. Here also the energy is not destroyed but is only redistributed from the positions of minimum intensity to those of maximum intensity. For, at the maximum intensity positions the intensity due to the two waves should be $2a^2$ but it is actually $4a^2$. As shown in Fig., the intensity varies from 0 to $4a^2$ and the average is still $2a^2$.



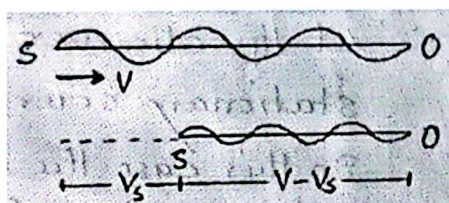
Doppler Effect

It is commonly observed that the frequency (or pitch) of a note apparently changes when either the source or the observer are in motion to each other. The apparent change in frequency due to a relative motion between the source and listener or observer is known as Doppler effect. This phenomenon was first investigated by C. Doppler in 1842.

Doppler effect in sound is asymmetric. When the source moves towards the observer with a certain velocity, the apparent pitch is different to the case when the observer is moving towards the source with same velocity. But it is not so in the case of light. Doppler effect in light is symmetric.

Case-1: Source in motion and observer at rest

(a) When the source moves towards the stationary observer:



Suppose a source S is producing sound of pitch (frequency) n and wavelength λ . The velocity of sound is V . Then, the wavelength $\lambda = \frac{V}{n}$

Let the source moves towards the observer O with a velocity V_s . In one second, n waves will be contained in a length $(V - V_s)$ and the apparent wavelength,

$$\lambda' = \frac{V - V_s}{n}$$

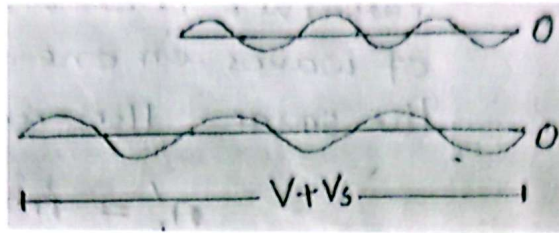
The apparent pitch heard by observer

$$n' = \frac{V}{\lambda'} = \left(\frac{V}{V - V_s} \right) n$$

$$\text{i.e., } n' > n$$

Thus, the apparent pitch of the note increases when the source moves towards a stationary observer.

(b) When the source moves away from the stationary observer:



Let the source moves away from the observer with a velocity V_s . In one second, n waves will be contained in a length $(V + V_s)$ and the apparent wavelength,

$$\lambda' = \frac{V + V_s}{n}$$

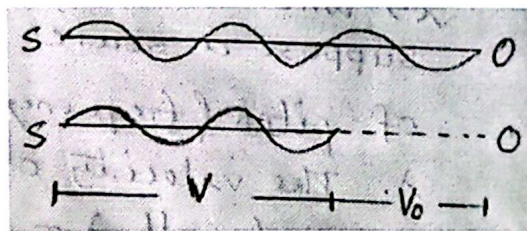
The apparent pitch, $n' = \frac{V}{\lambda'} = \left(\frac{V}{V + V_s} \right) n$

$$\text{i.e., } n' < n$$

Thus, the apparent pitch of the note decreases when the source moves away from a stationary observer.

Case-2: Source at rest and observer in motion

(a) When the observer moves towards a stationary source:



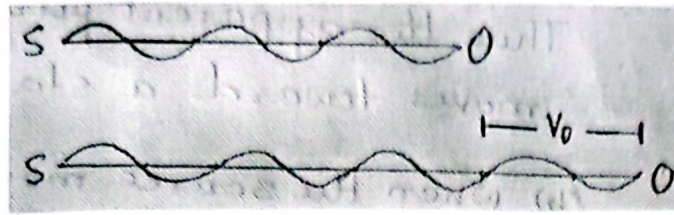
Let the observer moves towards a stationary source with a velocity V_o . In this case the observer receives more number of waves in one second. The apparent wave length remains the same. The apparent frequency,

$$\begin{aligned} n' &= n + \frac{V_o}{\lambda} = \frac{V}{\lambda} + \frac{V_o}{\lambda} \\ &= \frac{V + V_o}{\lambda} = \left(\frac{V + V_o}{V} \right) n \quad \left[\lambda = \frac{V}{n} \right] \end{aligned}$$

$$\text{i.e., } n' > n$$

Thus, the apparent pitch of the note increases when the observer moves towards the stationary source.

(b) When the observer moves away from a stationary source



Let the observer move away from a stationary source with a velocity V_o . In this case, the observer receives less number of waves in one second. The apparent wavelength remains the same. The apparent frequency,

$$n' = n - \frac{V_o}{\lambda} = \frac{V}{\lambda} - \frac{V_o}{\lambda}$$

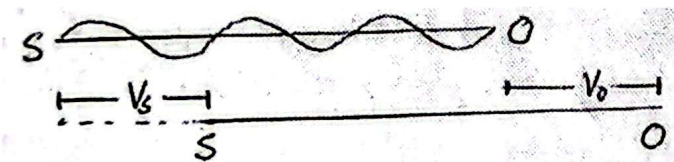
$$= \frac{V - V_o}{\lambda} = \left(\frac{V - V_o}{V} \right) n$$

i.e., $n' < n$

Thus, the apparent pitch of the note decreases when the observer moves away from a stationary source.

Case-3: Source and observer both in motion

When the source moves towards the observer and the observer moves away from the source:



Let the source move towards the observer with a velocity V_s and the observer moves away from the source with a velocity V_o . The apparent wavelength,

$$\lambda' = \frac{V - V_s}{n}$$

and apparent pitch, $n' = \frac{V - V_o}{\lambda'}$

$$= \left(\frac{V - V_o}{V - V_s} \right) n \dots \dots \dots (i)$$

Special Cases:

(a) When the source and observer move towards each other:

In equation (i), taking V_o to be negative $n' = \left[\frac{V - (-V_o)}{V - V_s} \right] n$

$$= \left(\frac{V + V_o}{V - V_s} \right) n$$

(b) When the source and observer move away from each other:

In equation (i), taking V_s to be negative

$$n' = \left[\frac{V - V_o}{V - (-V_s)} \right] n = \left(\frac{V - V_o}{V + V_s} \right) n$$

(c) When source moving away from the observer and the observer moving towards the source

In equation (i), taking both V_o and V_s negative

$$n' = \left[\frac{V - (-V_o)}{V - (-V_s)} \right] n = \left(\frac{V + V_o}{V + V_s} \right) n$$